Building a Custom Plant Care Tracker

Using Liferay Objects, Picklists, and Forms, you can keep track of all your plants, watering, and healthy status with a Custom Plant Care Tracker.

To build a Custom Plant Care Tracker, you will create two Picklists and configure them in the Liferay Portal. The Picklists enforce standardized data entry for fields like Plant Health and Environment, preventing inconsistent data and making the dashboard easier to filter. Next, you will create two Objects to store each plant's profile and care logs that share relational data. Separating the plant profile and care logs allows users to track watering schedules and health trends over time for each individual plant.

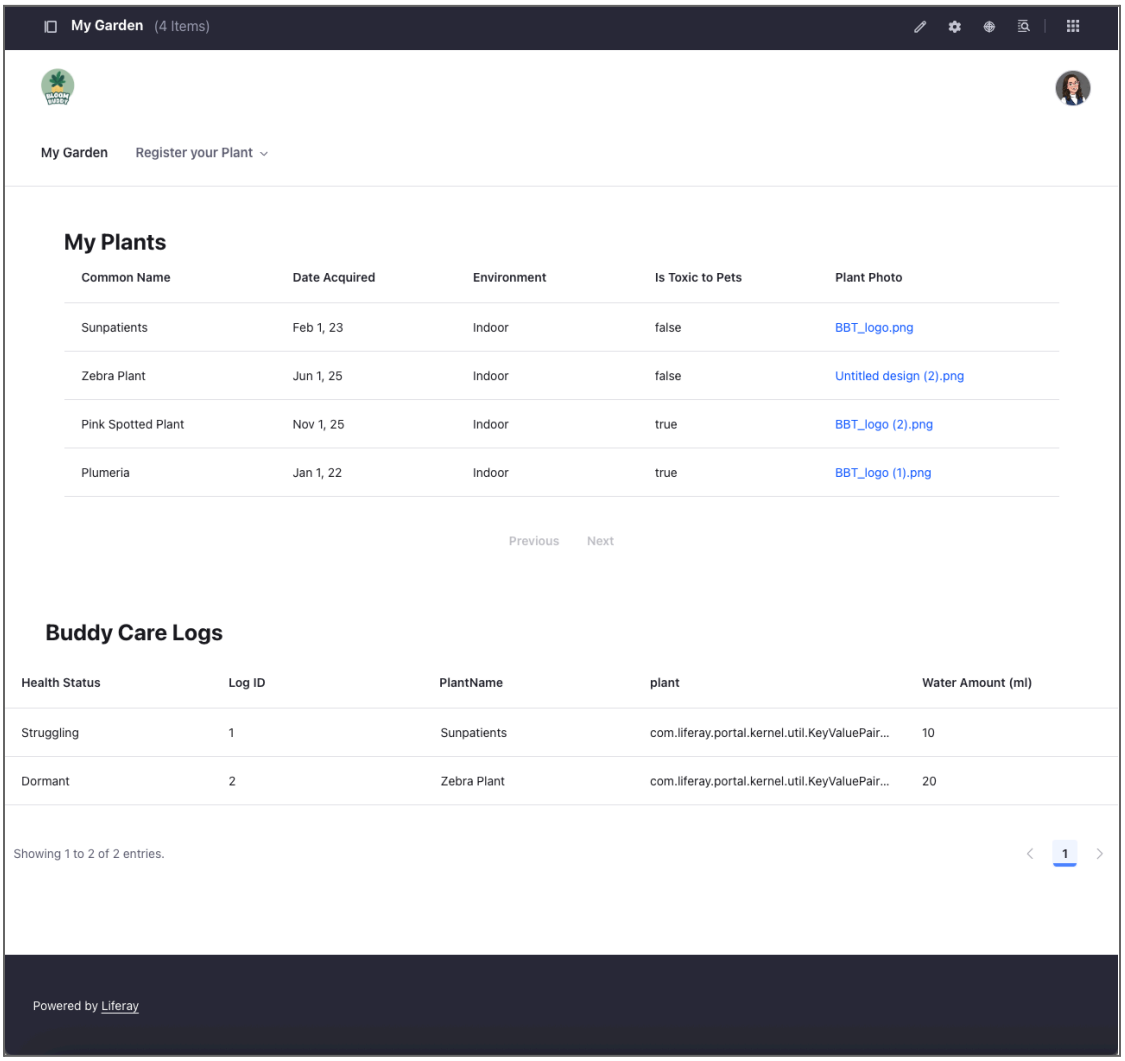


Table of Contents

Table of Contents 2
Prerequisites 2
Adding an Environment Picklist 2
Adding a Plant Health Picklist 4
Adding a Plant Object 5
Configuring a Plant Registration Page 7
Adding a Care Log Entry Page 9
Build the My Garden Dashboard 11
Verification (Testing) 13

Prerequisites

To follow this article, you must have:

- A hosting environment with a minimum allocation of **8GB of available RAM**.
- A running instance of **Liferay Portal 7.4 CE** (*GA100+*) by Installing a Liferay Tomcat Bundle and Configuring a Database
- A user account with **Administrative access** to the Global Menu and Site Menu.

Start a new **Liferay Portal CE** instance by running the following command in your terminal:

```
docker run -it -m 8g -p 8080:8080 liferay/portal:7.4.3.129-ga129
```

Sign in to Liferay at http://localhost:8080 using the email address 'test@liferay.com' and the password 'test'. When prompted, change the password.

Adding an Environment Picklist

The first step is to create two separate Objects to manage your data sets. The Plant Object stores core plant details, while the related Care Log Object tracks the ongoing health and watering history for each

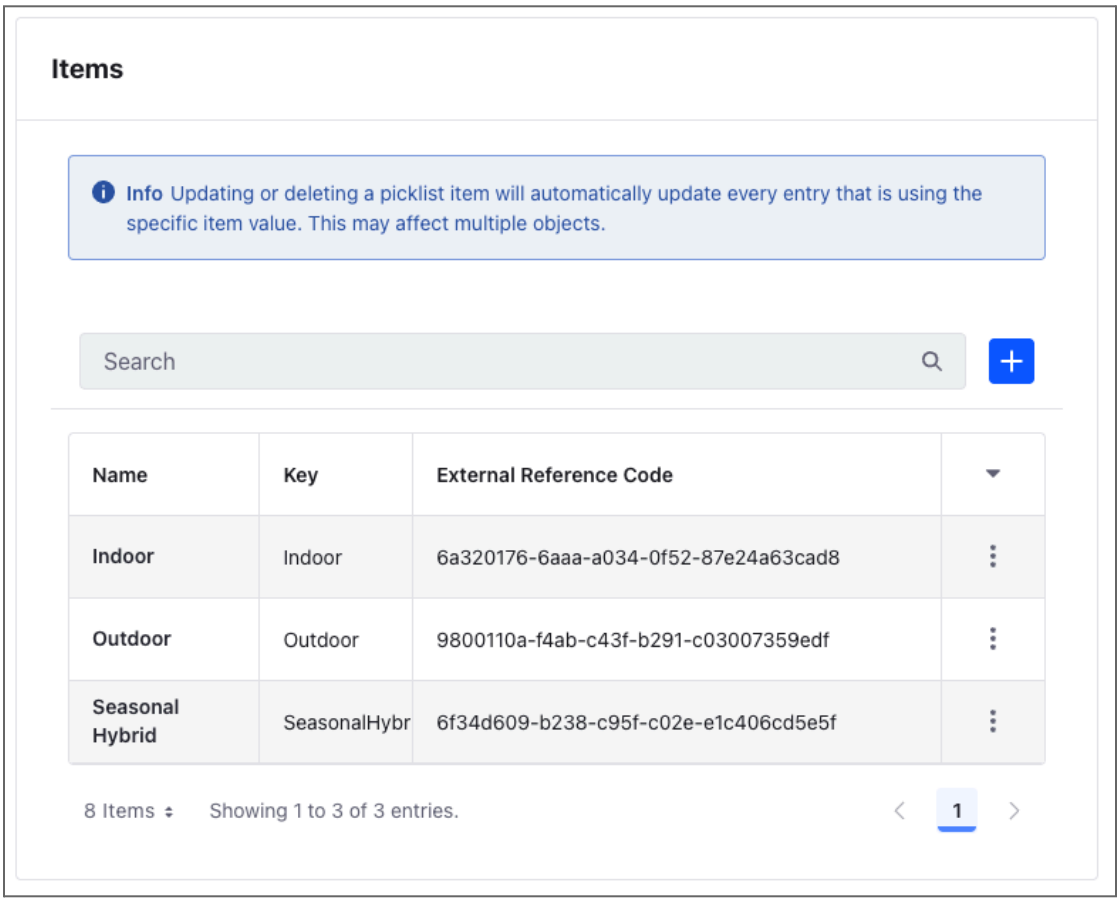
plant.

1. Open the *Global Menu* ( ), go to the *Control Panel* tab, and click ***Picklists***.

2. Click *Add* ( ), enter '*Environment*' for name, and then click ***Save***.

3. Select the new Picklist and use the Add button ( ) to add the following items:

Name	**Key**
Indoor	Indoor
Outdoor	Outdoor
Seasonal Hybrid	Seasonal Hybrid



You can now use the Environment Picklist as a field in the Plant Object.

Adding a Plant Health Picklist

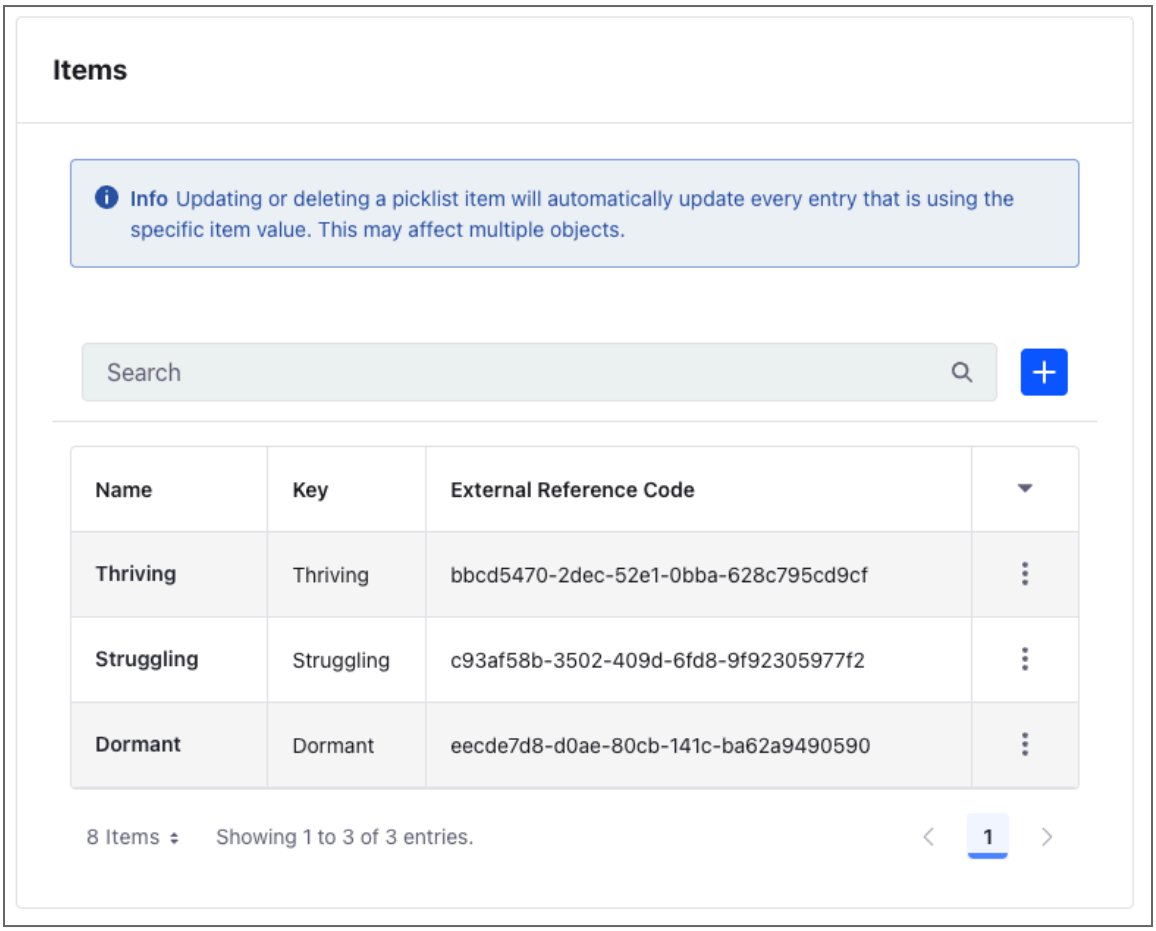
Next, configure the **Plant Health Picklist** to enforce standardized data entry for health assessments. Once created, this picklist is mapped to a field in the **Care Log Object,** ensuring that health status logs remain consistent across all plant entries.

1. Open the *Global Menu* ( ), go to the *Control Panel* tab, and click ***Picklists***.

2. Click *Add* ( ), enter '*Plant Health*' for name, and then click ***Save***.

3. Select the new Picklist and use the Add button ( ) to add the following items:

Name	**Key**
Thriving	Thriving
Struggling	Struggling
Dormant	Dormant



You can now use the Plant Health Picklist as a field in the Care Log Object.

Adding a Plant Object

The Plant Object stores the core attributes for each individual plant entry in the data set. As the parent entity in a one-to-many relationship, it provides the foundation for linking recurring care activities to individual records.

1. Open the *Global Menu* ( ), go to the *Control Panel* tab, and click ***Objects***.

2. Click *Add* ( ) and enter the following values:

Field	Value
Label	Plant
Plural Label	Plants
Object Name	Plant

3. Select the *Object*, click the *Fields* tab, and add the following fields:

Label	Field Name	Type	Picklist	Required
Common Name	commonName	Text	n/a	✓
Date Acquired	dateAcquired	Date	n/a	✓
Environment	environment	Picklist	Environment	✓
Plant Photo	plantPhoto	Attachment	n/a	
Is Toxic to Pets	isToxicToPets	Boolean	n/a	

Common Name	Text	Yes	CUSTOM	No	⋮
Date Acquired	Date	Yes	CUSTOM	No	⋮
Environment	Picklist	Yes	CUSTOM	No	⋮
Plant Photo	Attachment	No	CUSTOM	No	⋮
Is Toxic to Pets	Boolean	No	CUSTOM	No	⋮

NOTE: The Plant Object fields are customizable and are fully extensible beyond the data attributes described in this article.

Adding a Care Log Object

Creating a Care Log Object allows users to record maintenance activities, such as watering schedules and health assessments. The transactional data linked in the Care Log Object is specific to an individual plant's care history.

1. Open the *Global Menu* ( ), go to the *Control Panel* tab, and click ***Objects***.

2. Click *Add* ( ) and enter the following values:

Field	**Value**
Label	Care Log
Plural Label	Care Logs
Object Name	CareLog

3. Select the *Object*, click the *Fields* tab, and add the following fields:

Label	**Field Name**	**Type**	**Picklist**	**Required**
Log ID	logID	Auto Increment	n/a	✓
Water Amount (ml)	waterAmount	Integer	n/a	✓
Health Status	healthStatus	Picklist	Plant Health	✓
plant	r_plant_c_plantId	Relationship	n/a	✓

Log ID	Auto Increment	No	CUSTOM	No	⋮
Water Amount (ml)	Integer	Yes	CUSTOM	No	⋮
Health Status	Picklist	Yes	CUSTOM	No	⋮
plant	Relationship	Yes	CUSTOM	No	

Now that the data set Objects and Picklists are configured, the next step is to create an intake form for users to register their plants.

Configuring a Plant Registration Page

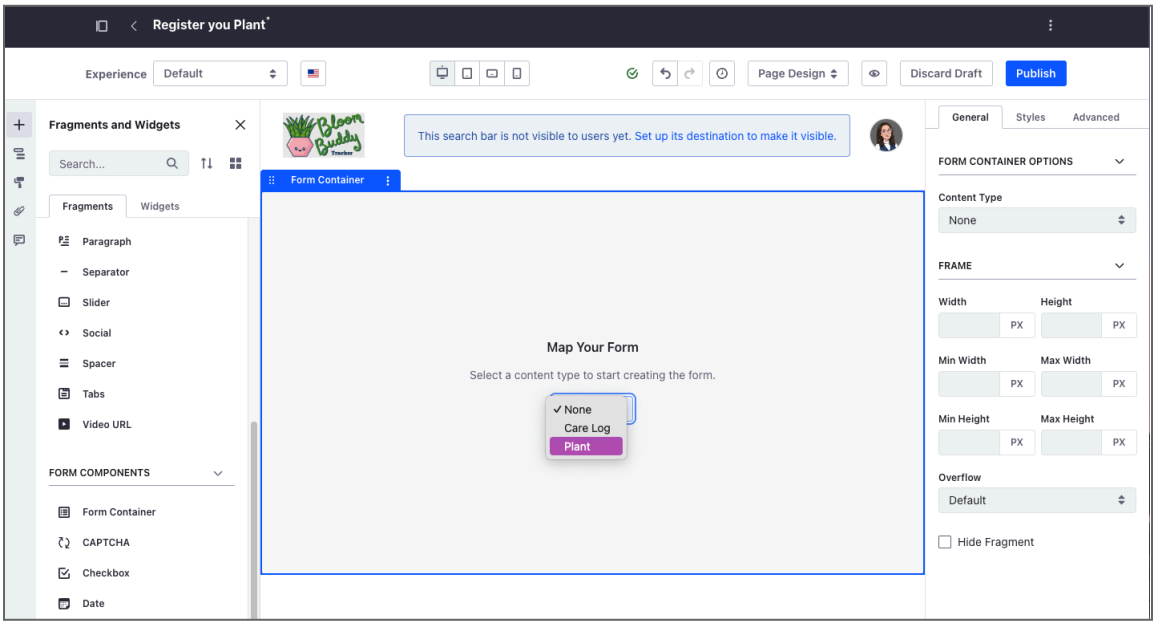
The Plant Object Fields are added to a Form Container Fragment and mapped to create the Plant Registration page. On the Plant Registration page, users can add all the details about their plants and log data in the Plant Object.

1. Build a New Site.

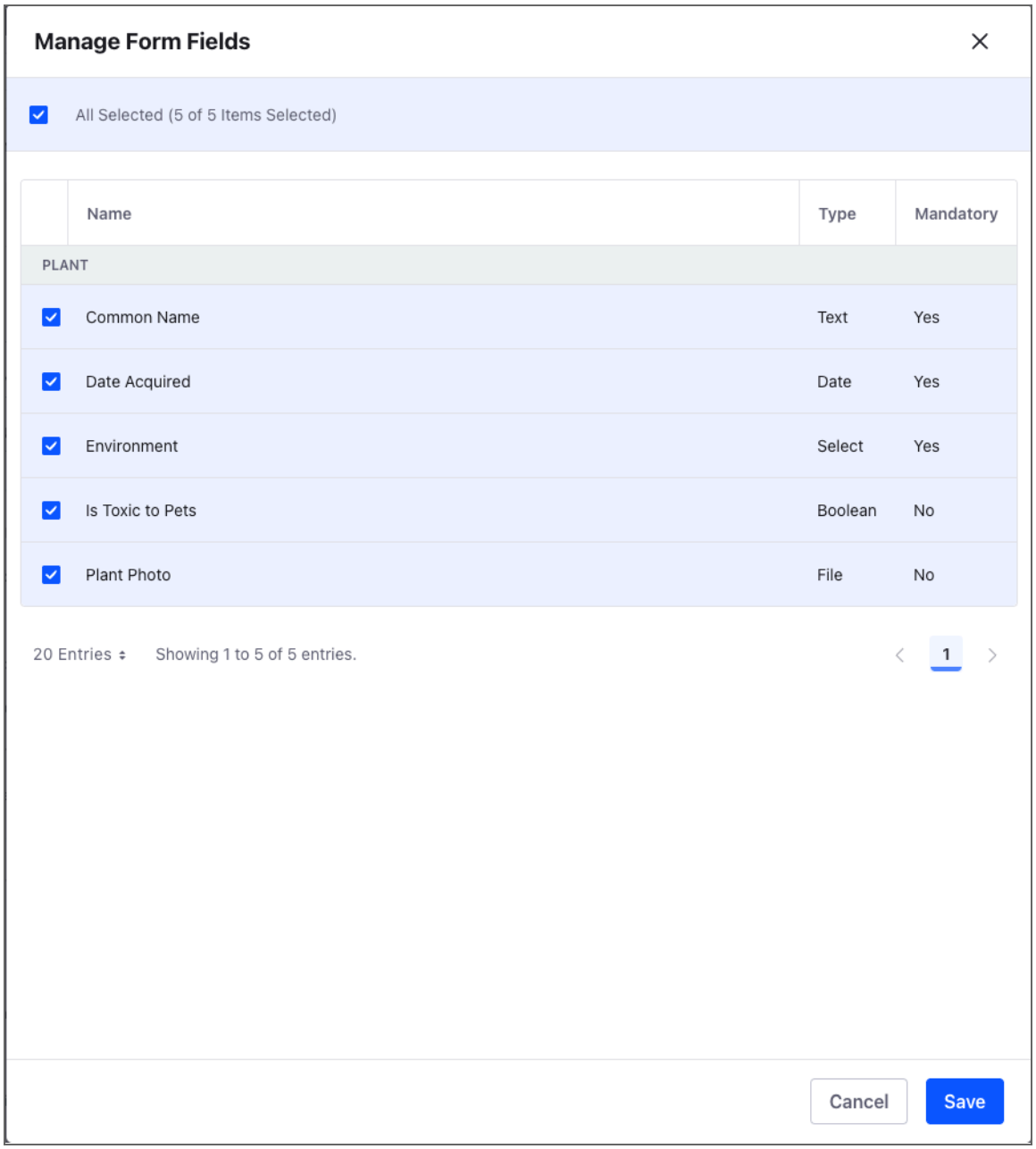
2. In the Site Editor, add a New Page.

3. From the left-sidebar menu in the page editor, click and drag ***Form Container*** to the page.

4. Click the drop-down menu, and then select ***Plant***.



5. Select all the 'Plant' form fields, and then click ***Save***.



6. Click ***Publish***.

Adding a Care Log Entry Page

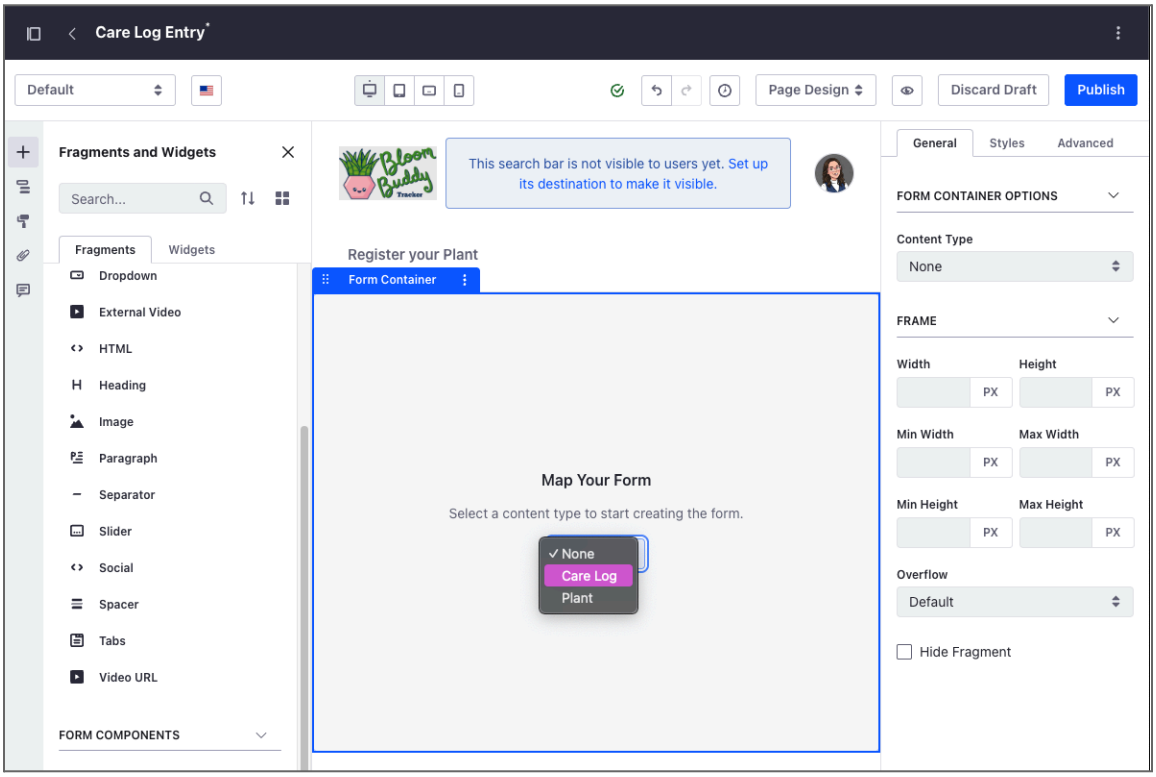
The **relational data** model between the Plant and Care Log Objects allows for dynamic data selection. By defining this relationship before building the Care Log form, the 'plant' field appears as a dropdown menu. This allows users to associate specific care routines with existing entries from the Plant

Registration page, ensuring data integrity across the application.

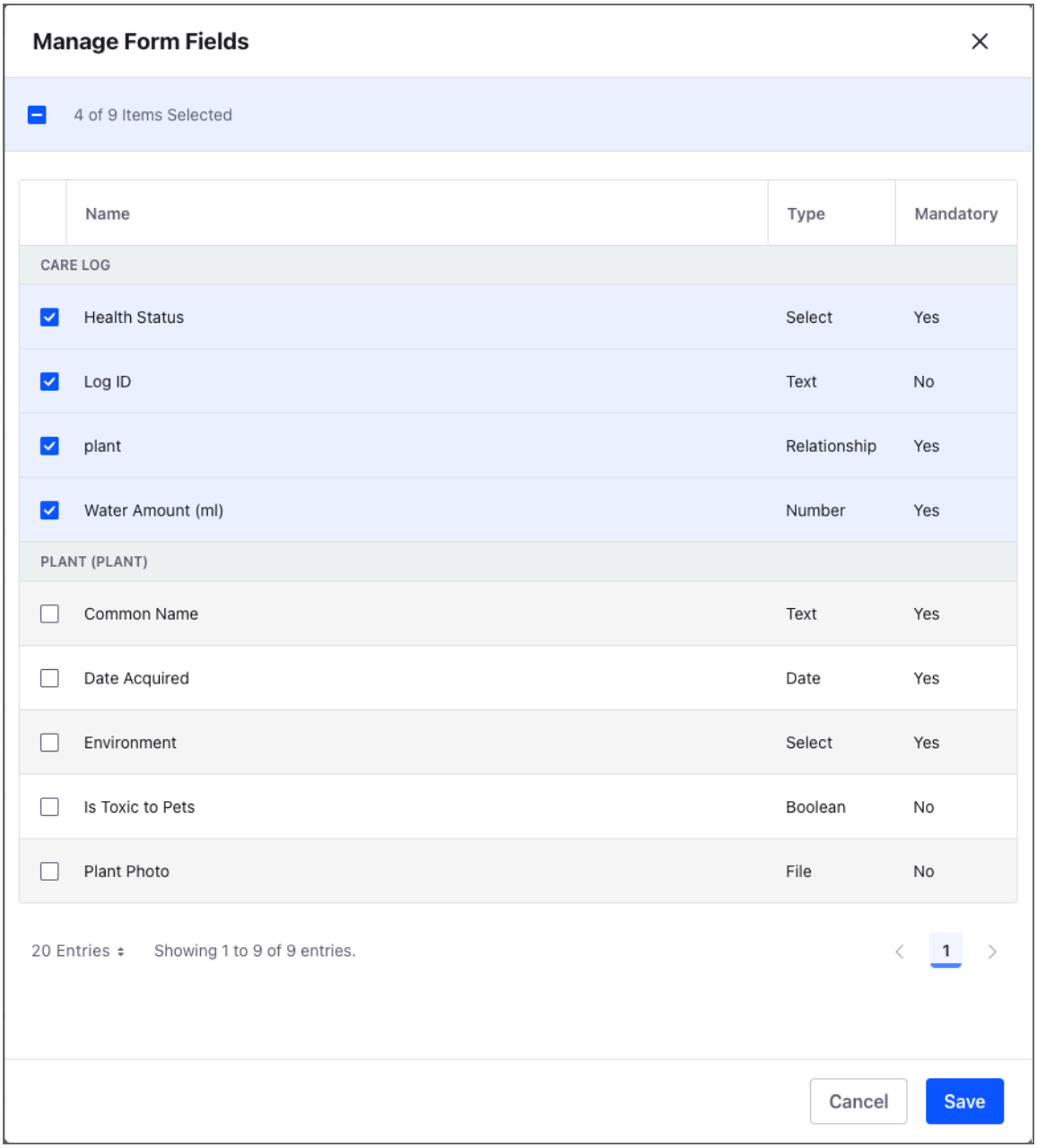
1. In the Site Editor, add a New Page.

2. From the left-sidebar menu in the page editor, click and drag ***Form Container*** to the page.

3. Click the drop-down menu, and then select ***Care Log***.



4. Select all the '*Care Log*' form fields, and then click ***Save***.



5. Click ***Publish***.

Build the My Garden Dashboard

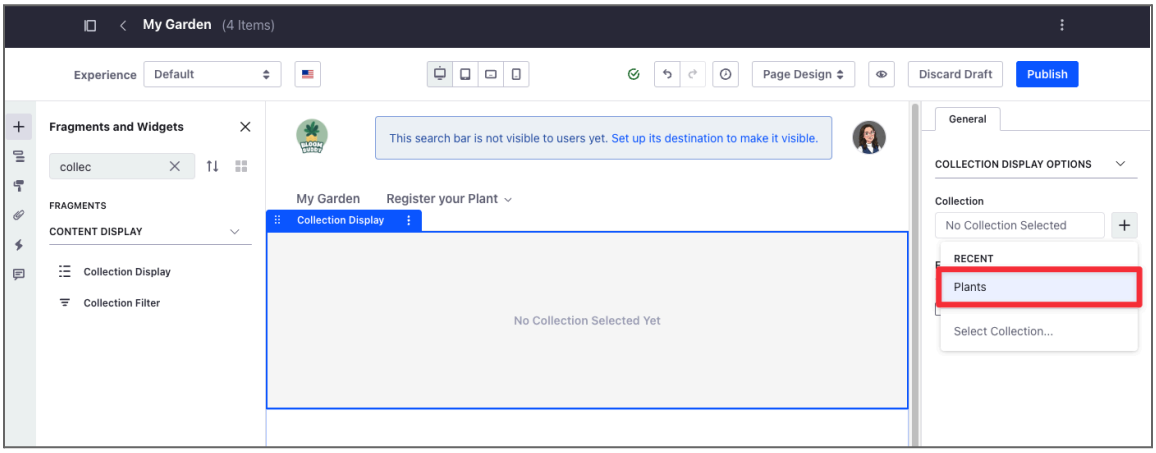
Lastly, you must configure a dashboard to visualize your registered plants alongside their associated care logs. Using [Collection Pages and Collection Display Fragments](), you can build a dedicated

dashboard to provide visibility into your stored datasets. This dashboard displays the mapped entries from both Objects.

1. In the Site Editor, add a New Page.

2. From the left-sidebar menu in the page editor, click and drag ***Collection Display*** to the page.

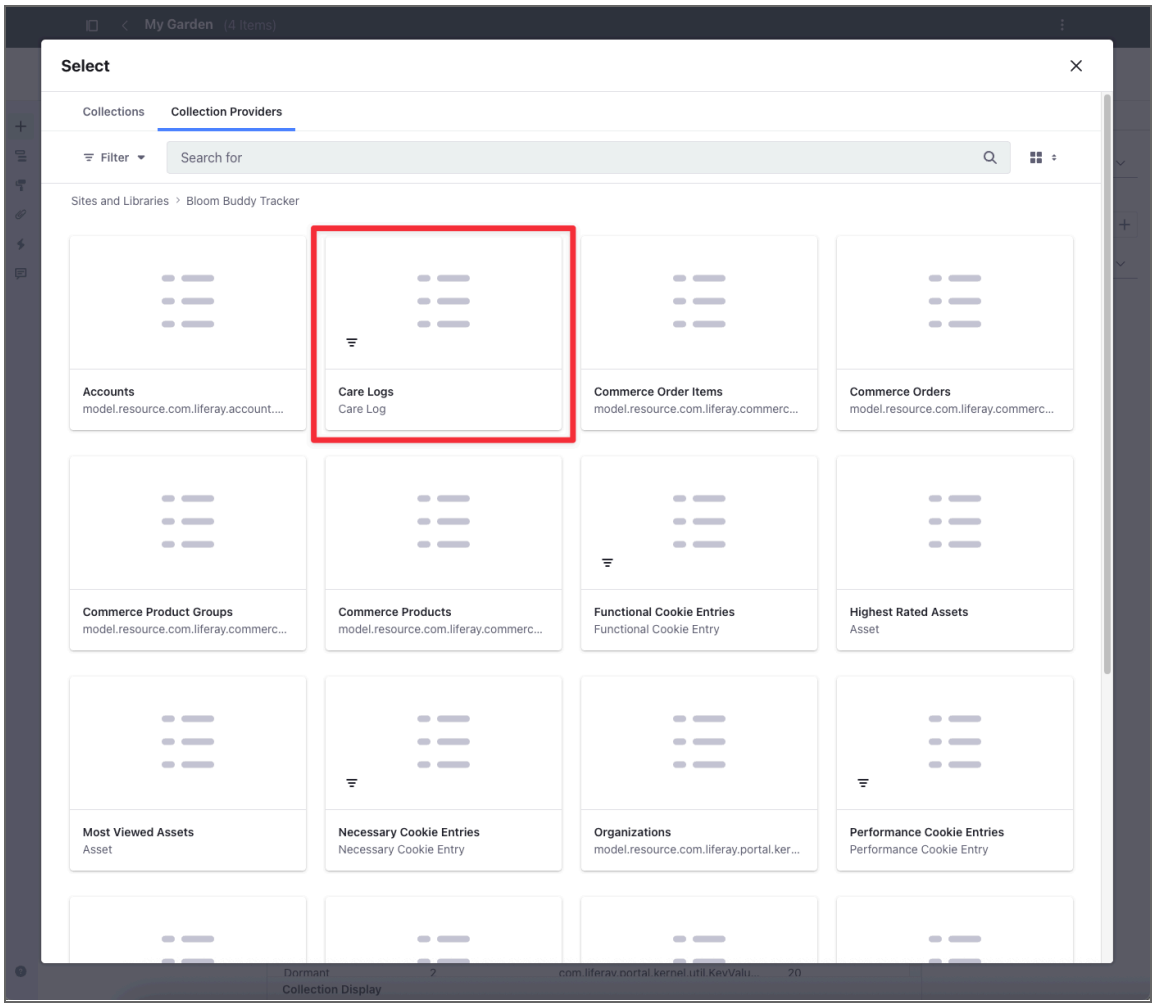
3. In the General tab, open the *Collection Selection* (+), and then click ***Plants***.



4. Select ***Table*** from the *Style Display* dropdown menu.

5. Add another *Collection Display Fragment***,** and then click ***Select Collection***.

6. Click *Collection Providers*, and then click ***Care Logs Collection***.



7. Select ***Table*** from the *Style Display* dropdown menu.

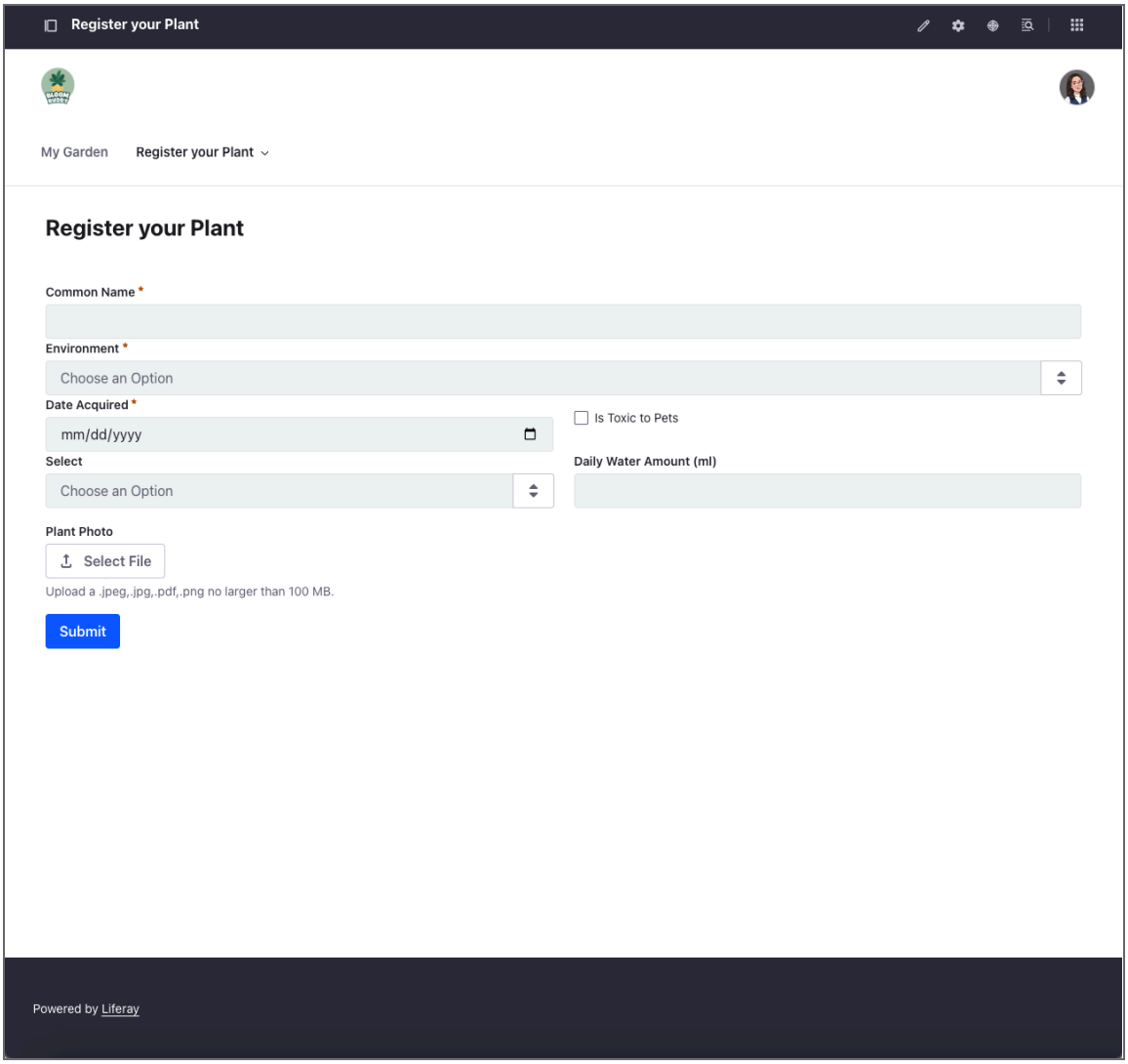
8. Click ***Publish*** to create the My Garden Dashboard.

Verification (Testing)

You have configured Objects, Picklists, Forms, and used Collection Display Fragments to create the Plant Care Tracker dashboard. The last step is to verify that the configurations are working properly after setup.

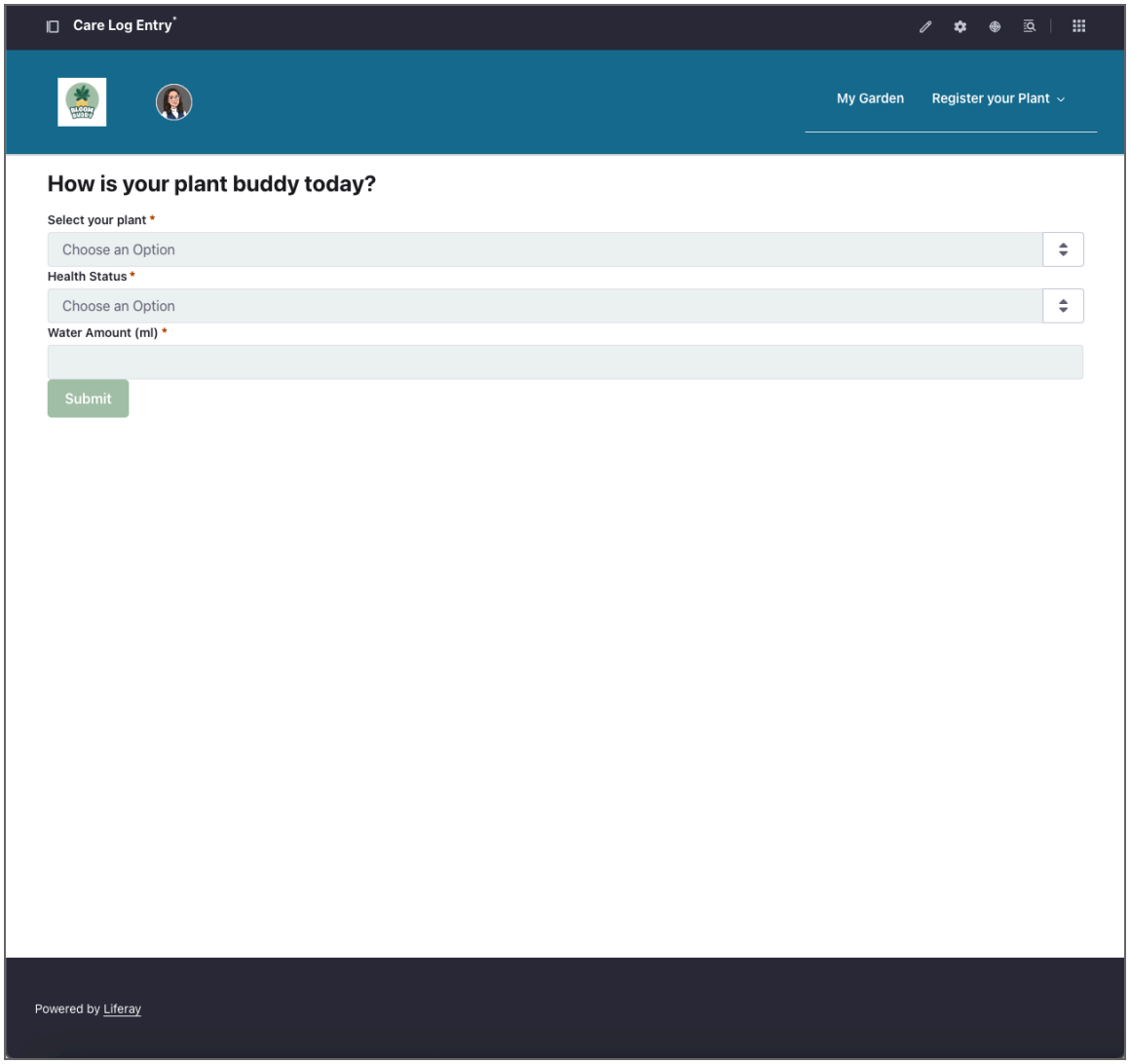
To verify that your Custom Plant Care Tracker is working

1. Visit your site's ***Register Your Plant*** page.

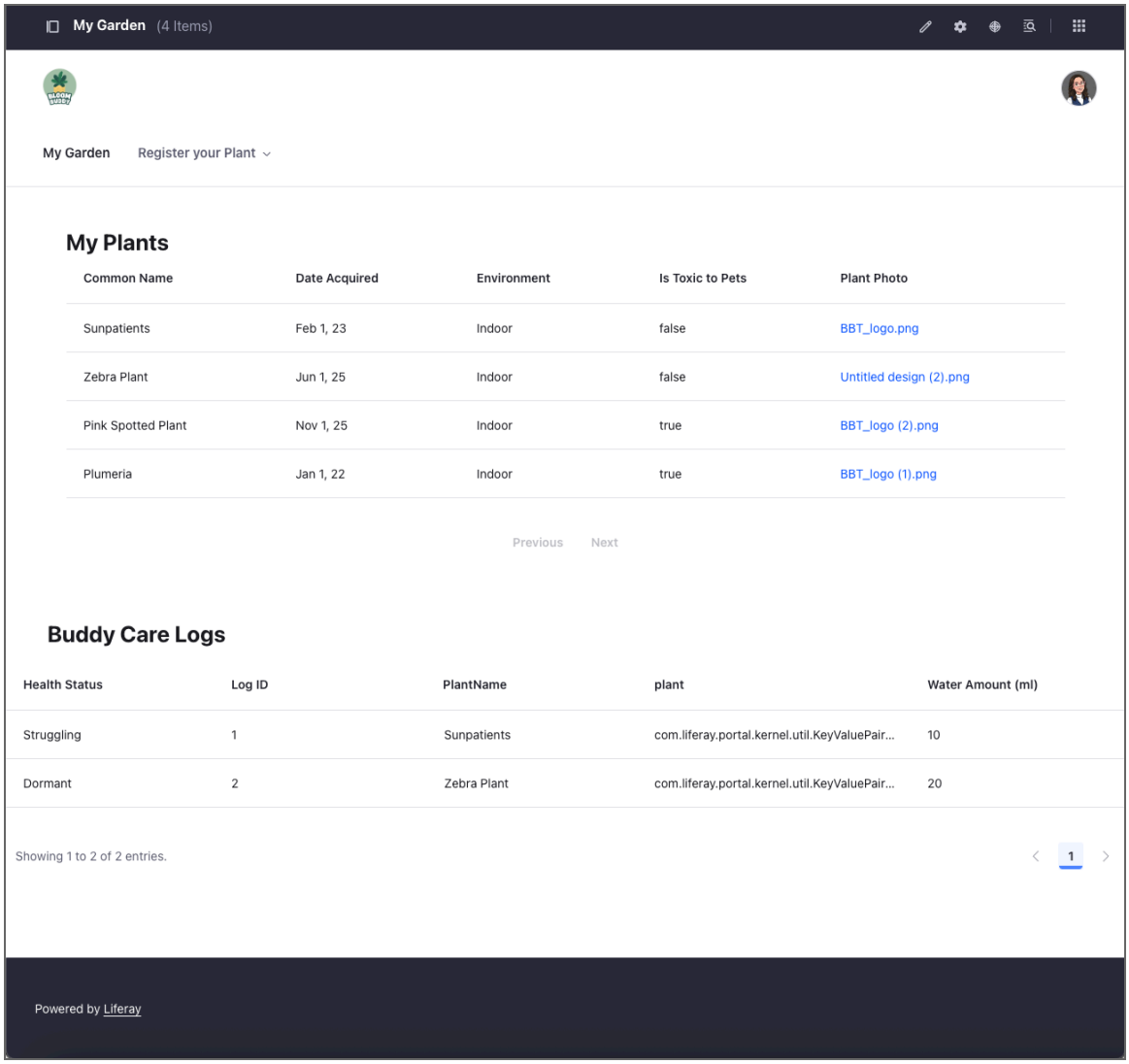


2. Fill out the *Register your Plant* form, and then click ***Submit***.

3. Visit your *Care Log Entry* page. Complete the form, and then click ***Submit***.



Upon successful form submission, the ***My Garden*** page dynamically retrieves the stored Object data. Both entries are displayed within the collection, confirming that the **standardized data entry** and field mappings are functioning as intended.


My Garden (4 Items)
My Garden
Register your Plant
My Plants
Common Name
Date Acquired
Environment
Is Toxic to Pets
Plant Photo
Sunpatients
Feb 1, 23
Indoor
false
BBT_logo.png
Zebra Plant
Jun 1, 25
Indoor
false
Untitled design (2).png
Pink Spotted Plant
Nov 1, 25
Indoor
true
BBT_logo (2).png
Plumeria
Jan 1, 22
Indoor
true
BBT_logo (1).png
Previous
Next
Buddy Care Logs
Health Status
Log ID
PlantName
plant
Water Amount (ml)
Struggling
1
Sunpatients
com.liferay.portal.kernel.util.KeyValuePair...
10
Dormant
2
Zebra Plant
com.liferay.portal.kernel.util.KeyValuePair...
20
Showing 1 to 2 of 2 entries.
1
Powered by Liferay